

```
import pandas as pd
import numpy as np

# Display all columns
pd.set_option('display.max_columns', None)
```

```
from google.colab import files
uploaded = files.upload()
```

Choose files aerofit_treadmill.csv
aerofit_treadmill.csv(text/csv) - 7279 bytes, last modified: 28/07/2026 - 100% done
 Saving aerofit_treadmill.csv to aerofit_treadmill.csv

```
import pandas as pd

df = pd.read_csv("aerofit_treadmill.csv")
```

Problem Statement

Aerofit is a fitness equipment company that sells different treadmill models.

The objective is to analyze customer characteristics such as age, gender, education, marital status, income, fitness level, usage frequency, and miles walked to understand purchasing behavior across different treadmill models.

The analysis will help Aerofit identify customer profiles for each product and support data-driven business decisions.

Shape of Dataset

```
df.shape
```

```
(180, 9)
```

Dataset Information

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 180 entries, 0 to 179
Data columns (total 9 columns):
 #   Column          Non-Null Count  Dtype
---  -
 0   Product         180 non-null   object
 1   Age             180 non-null   int64
 2   Gender          180 non-null   object
 3   Education       180 non-null   int64
 4   MaritalStatus   180 non-null   object
 5   Usage           180 non-null   int64
 6   Fitness         180 non-null   int64
 7   Income          180 non-null   int64
 8   Miles           180 non-null   int64
dtypes: int64(6), object(3)
memory usage: 12.8+ KB
```

Convert Categorical Variables

```
categorical_cols = ['Product', 'Gender', 'MaritalStatus']

for col in categorical_cols:
    df[col] = df[col].astype('category')
```

Check Missing Values

```
df.isnull().sum()
```

	0
Product	0
Age	0
Gender	0
Education	0
MaritalStatus	0
Usage	0
Fitness	0
Income	0
Miles	0

dtype: int64

Check Duplicate Records

df.duplicated().sum()

np.int64(0)

Statistical Summary

df.describe(include='all').T

	count	unique	top	freq	mean	std	min	25%	50%	75%	max	
Product	180	3	KP281	80	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
Age	180.0	NaN	NaN	NaN	28.788889	6.943498	18.0	24.0	26.0	33.0	50.0	
Gender	180	2	Male	104	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
Education	180.0	NaN	NaN	NaN	15.572222	1.617055	12.0	14.0	16.0	16.0	21.0	
MaritalStatus	180	2	Partnered	107	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
Usage	180.0	NaN	NaN	NaN	3.455556	1.084797	2.0	3.0	3.0	4.0	7.0	
Fitness	180.0	NaN	NaN	NaN	3.311111	0.958869	1.0	3.0	3.0	4.0	5.0	
Income	180.0	NaN	NaN	NaN	53719.577778	16506.684226	29562.0	44058.75	50596.5	58668.0	104581.0	
Miles	180.0	NaN	NaN	NaN	103.194444	51.863605	21.0	66.0	94.0	114.75	360.0	

Observations

Dataset contains 180 observations and 9 variables.

There are 6 numerical and 3 categorical variables.

No missing values were found.

No duplicate records exist.

Categorical variables were converted to the category datatype for efficient memory usage.

Customers belong to a wide age range (18–50 years), indicating that treadmills appeal to both young and middle-aged adults.

Income shows substantial variation, suggesting the presence of multiple purchasing segments.

Weekly treadmill usage ranges from 2 to 7 sessions, while expected weekly miles range from 21 to 360 miles, indicating varying fitness goals.

The dataset is clean and well-structured, making it suitable for further exploratory data analysis and statistical modeling.

Display Number of Unique Values

df.nunique()

	0
Product	3
Age	32
Gender	2
Education	8
MaritalStatus	2
Usage	6
Fitness	5
Income	62
Miles	37

dtype: int64

Display Unique Values of Each Column

```
for col in df.columns:
    print(f"\n{col}")
    print(df[col].unique())
```

```
Product
['KP281', 'KP481', 'KP781']
Categories (3, object): ['KP281', 'KP481', 'KP781']
```

```
Age
[18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41
 43 44 46 47 50 45 48 42]
```

```
Gender
['Male', 'Female']
Categories (2, object): ['Female', 'Male']
```

```
Education
[14 15 12 13 16 18 20 21]
```

```
MaritalStatus
['Single', 'Partnered']
Categories (2, object): ['Partnered', 'Single']
```

```
Usage
[3 2 4 5 6 7]
```

```
Fitness
[4 3 2 1 5]
```

```
Income
[ 29562  31836  30699  32973  35247  37521  36384  38658  40932  34110
  39795  42069  44343  45480  46617  48891  53439  43206  52302  51165
  50028  54576  68220  55713  60261  67083  56850  59124  61398  57987
  64809  47754  65220  62535  48658  54781  48556  58516  53536  61006
  57271  52291  49801  62251  64741  70966  75946  74701  69721  83416
  88396  90886  92131  77191  52290  85906  103336  99601  89641  95866
 104581  95508]
```

```
Miles
[112  75  66  85  47 141 103  94 113  38 188  56 132 169  64  53 106  95
 212  42 127  74 170  21 120 200 140 100  80 160 180 240 150 300 280 260
 360]
```

Value Counts

```
categorical_cols = ['Product', 'Gender', 'MaritalStatus']

for col in categorical_cols:
    print(f"\nValue Counts of {col}")
    print(df[col].value_counts())
```

```
Value Counts of Product
Product
KP281    80
KP481    60
KP781    40
Name: count, dtype: int64
```

```
Value Counts of Gender
Gender
```

```

Male      104
Female    76
Name: count, dtype: int64

Value Counts of MaritalStatus
MaritalStatus
Partnered    107
Single       73
Name: count, dtype: int64

```

Percentage Distribution

```

for col in categorical_cols:
    print(f"\nPercentage Distribution of {col}")
    print(round(df[col].value_counts(normalize=True)*100,2))

```

```

Percentage Distribution of Product
Product
KP281    44.44
KP481    33.33
KP781    22.22
Name: proportion, dtype: float64

```

```

Percentage Distribution of Gender
Gender
Male      57.78
Female    42.22
Name: proportion, dtype: float64

```

```

Percentage Distribution of MaritalStatus
MaritalStatus
Partnered    59.44
Single       40.56
Name: proportion, dtype: float64

```

Non-Graphical Analysis

The dataset consists of 3 treadmill models (KP281, KP481, KP781). KP281 is the most frequently purchased model, indicating its popularity among customers.

There are 2 gender categories, with males constituting a slightly higher proportion than females.

Most customers are partnered, suggesting that treadmill purchases may be more common among individuals with partners.

Income has the highest number of unique values, showing significant variation in customers' earning levels.

Age also exhibits a wide range of values, indicating that Aerofit products appeal to customers across different age groups.

The categorical variables contain only a few unique values

```

import matplotlib.pyplot as plt
import seaborn as sns

sns.set_style("whitegrid")
plt.rcParams["figure.figsize"] = (8,5)

```

Univariate Analysis

```

numerical_cols = ['Age', 'Education', 'Usage', 'Fitness', 'Income', 'Miles']
numerical_cols

```

```

['Age', 'Education', 'Usage', 'Fitness', 'Income', 'Miles']

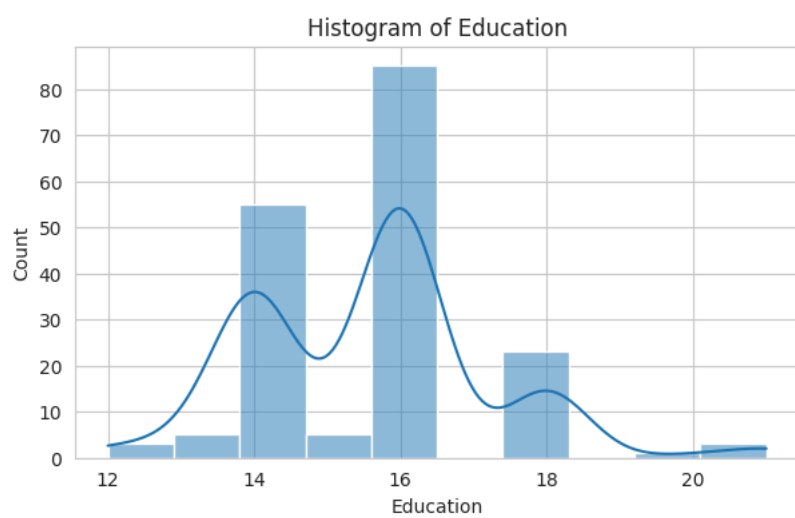
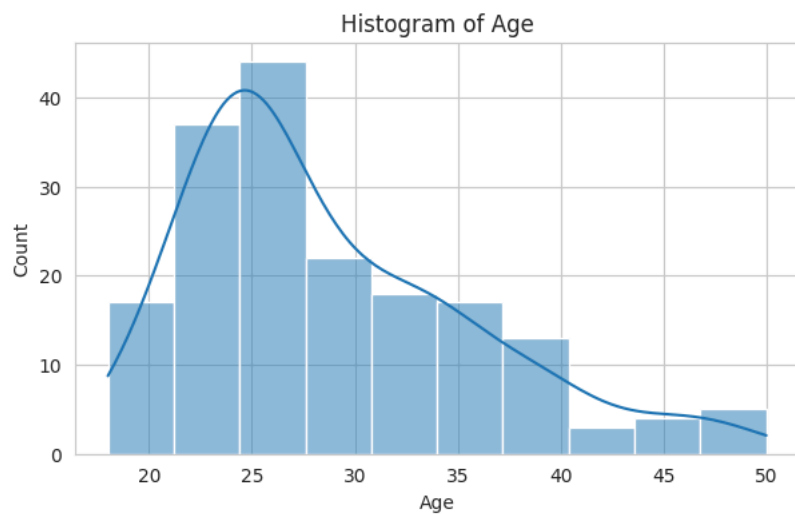
```

Histograms

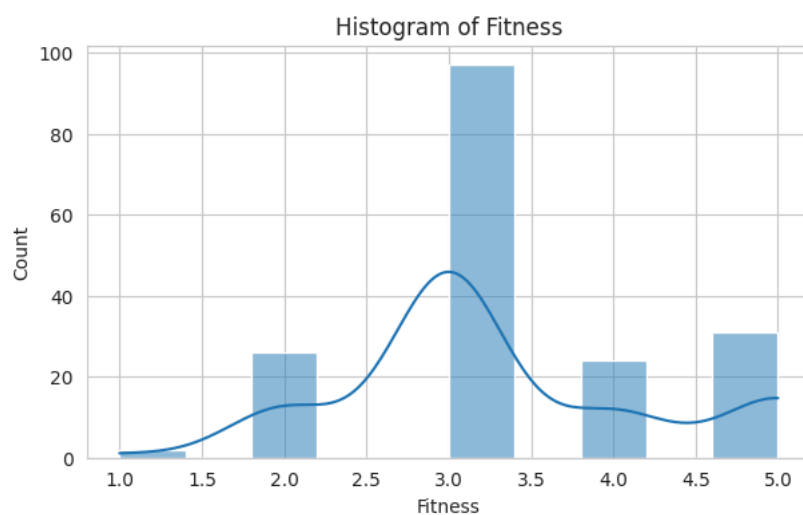
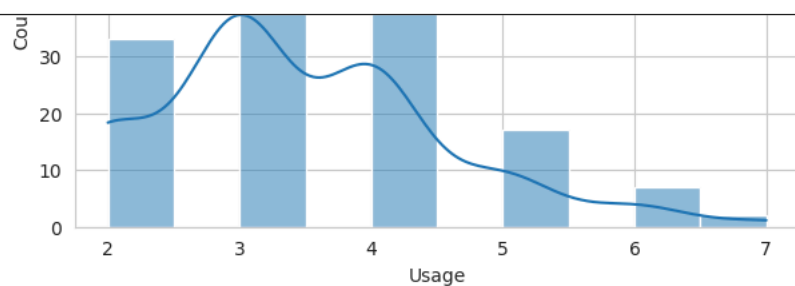
```

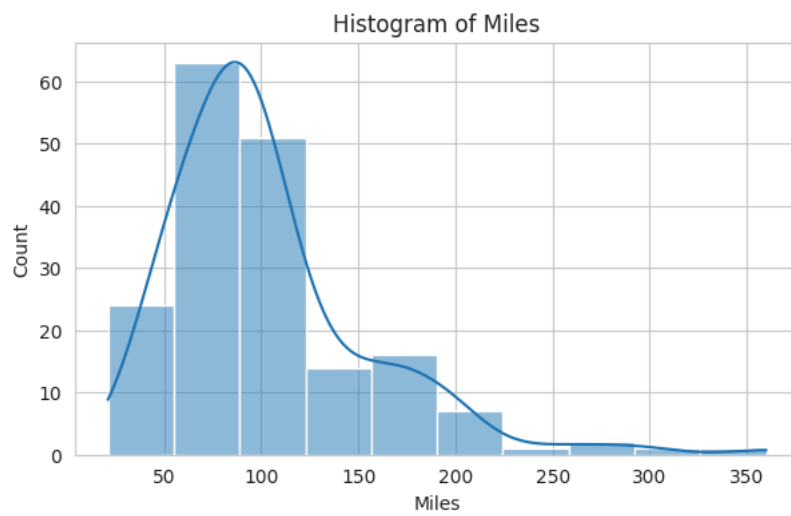
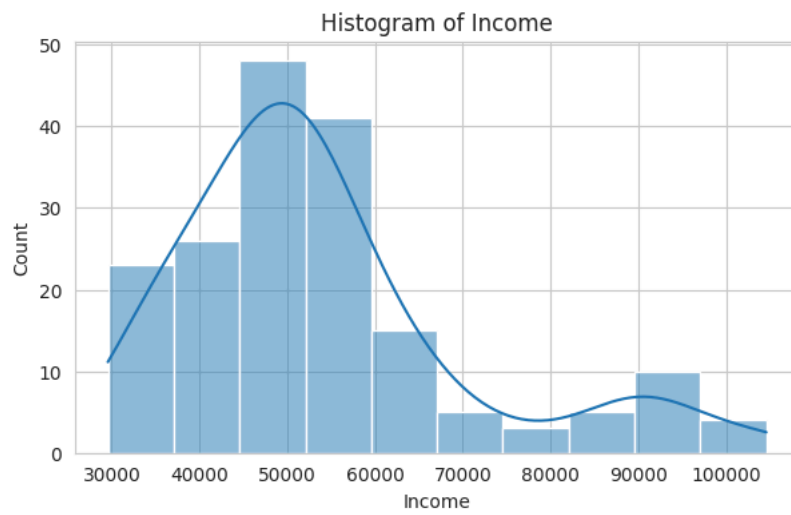
for col in numerical_cols:
    plt.figure(figsize=(7,4))
    sns.histplot(df[col], bins=10, kde=True)
    plt.title(f'Histogram of {col}')
    plt.show()

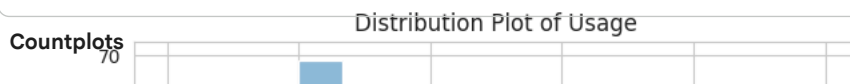
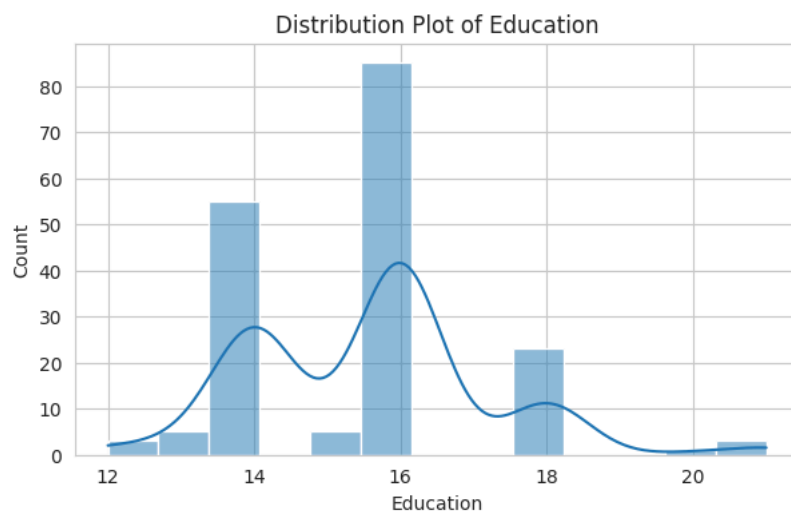
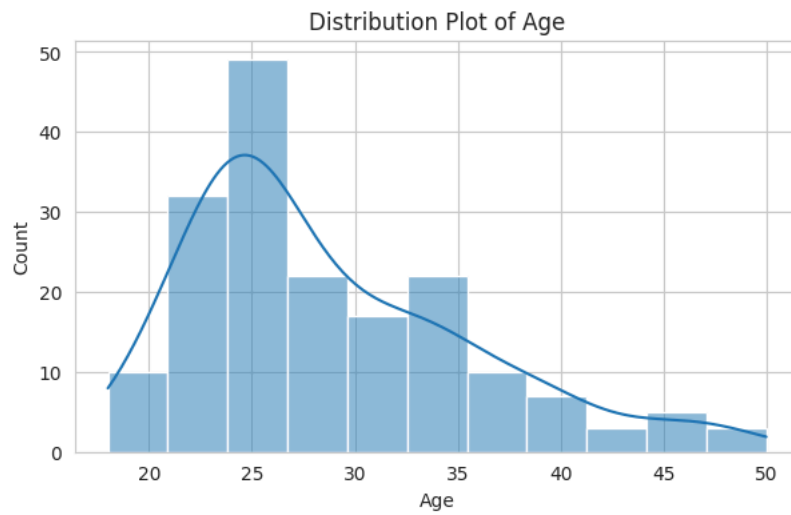
```

```
for col in numerical_cols:
    plt.figure(figsize=(7,4))
    sns.histplot(df[col], kde=True)
    plt.title(f'Distribution Plot of {col}')
    plt.show()
```

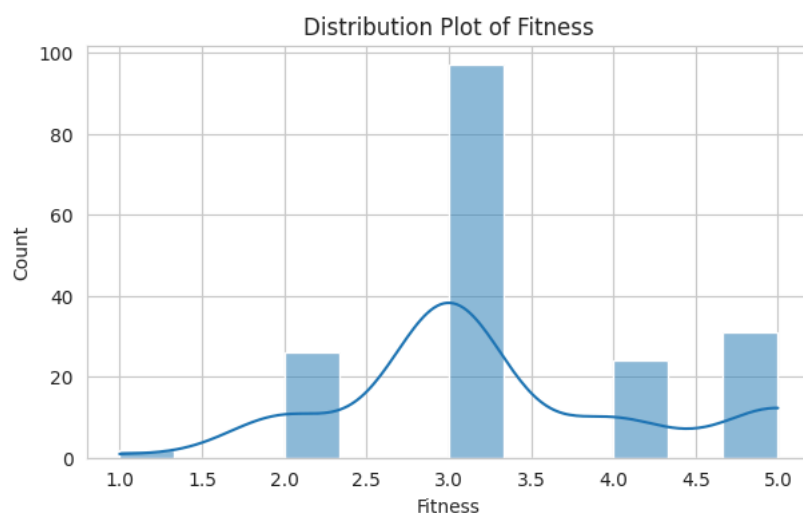
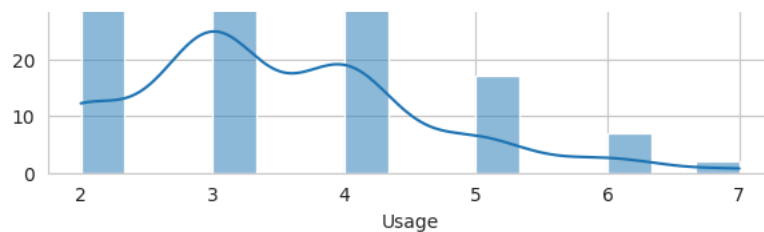


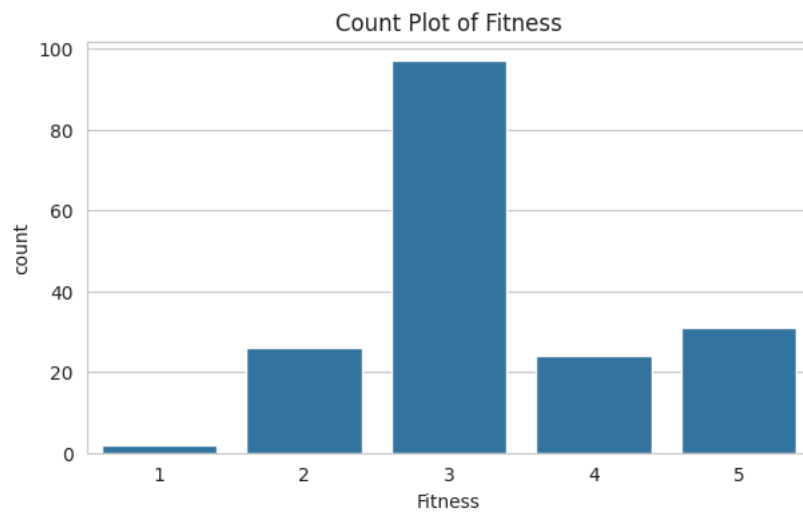
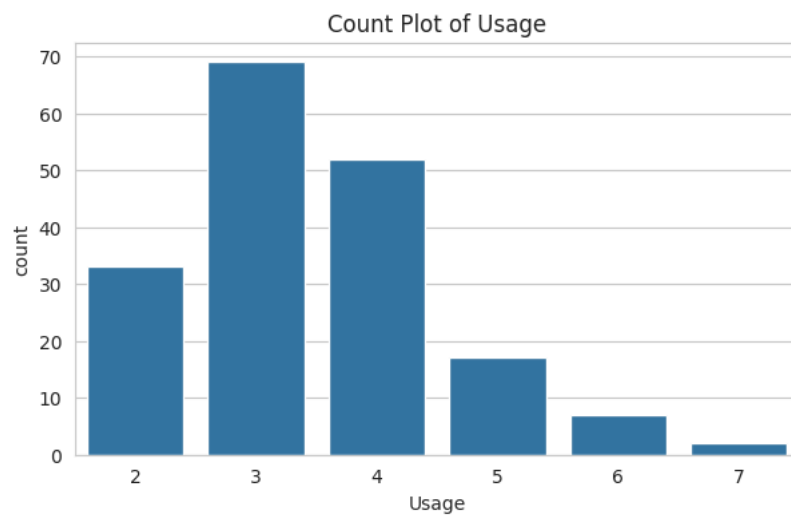
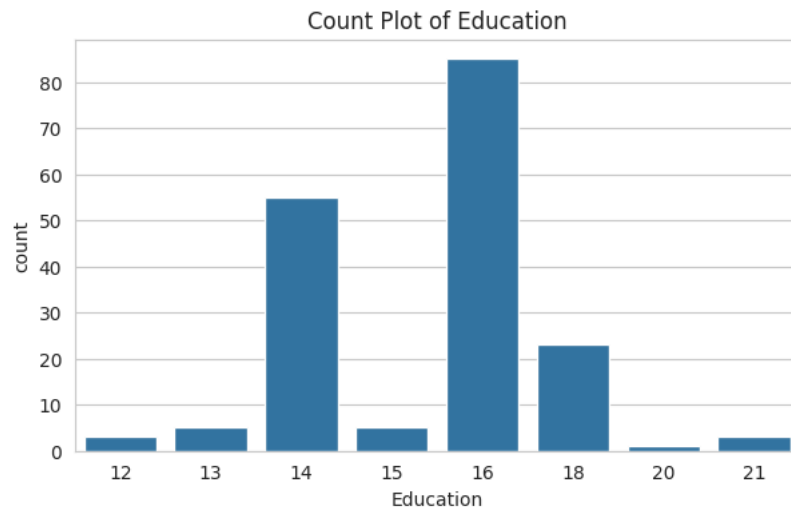




```
discrete_cols = ['Education', 'Usage', 'Fitness']
```

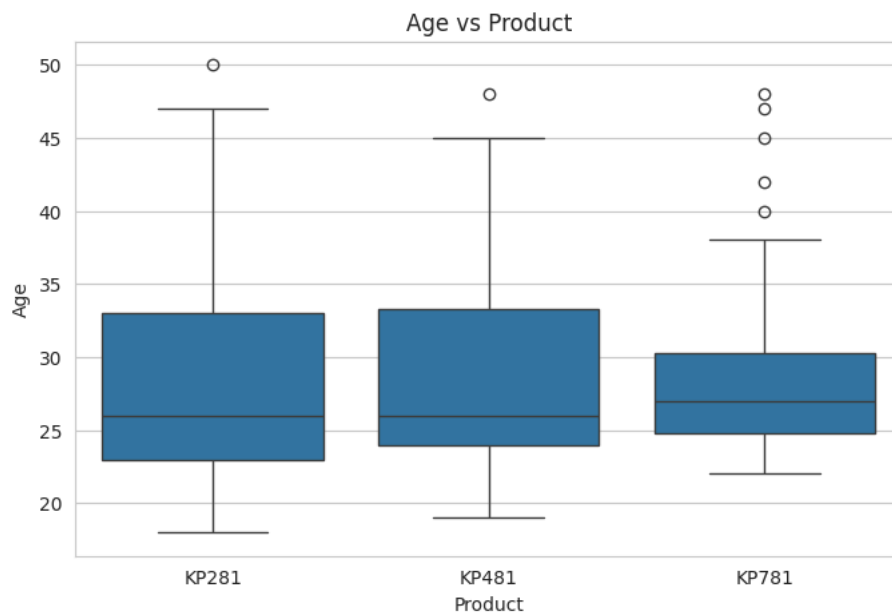
```
for col in discrete_cols:
    plt.figure(figsize=(7,4))
    sns.countplot(x=df[col])
    plt.title(f'Count Plot of {col}')
    plt.show()
```





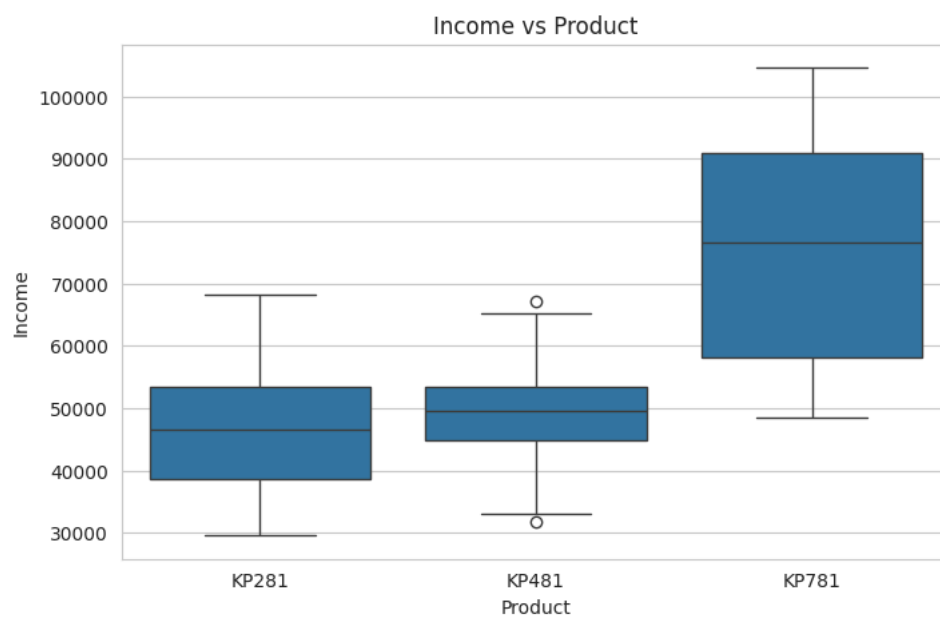
Categorical Variables using Boxplots

```
plt.figure(figsize=(8,5))
sns.boxplot(x='Product', y='Age', data=df)
plt.title("Age vs Product")
plt.show()
```



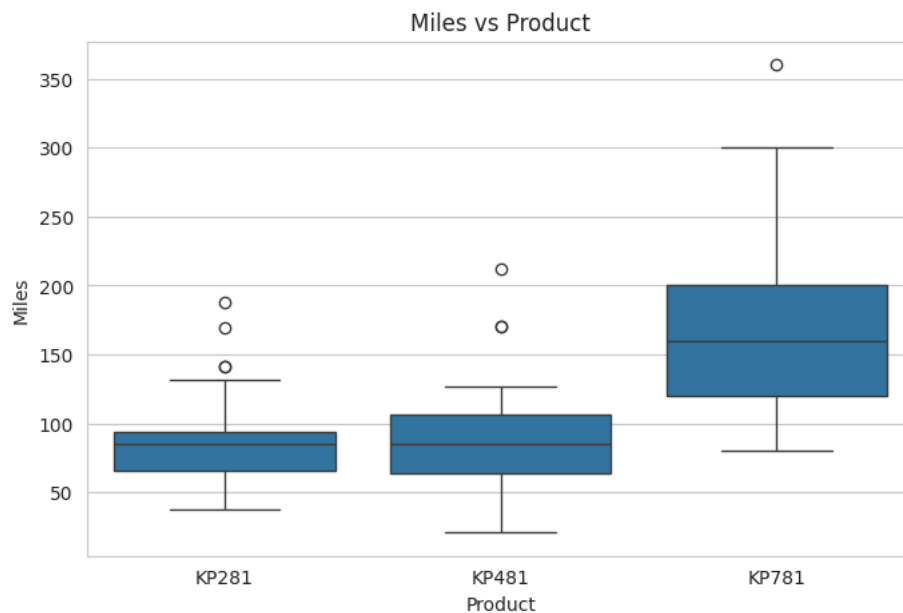
Income vs Product

```
plt.figure(figsize=(8,5))
sns.boxplot(x='Product', y='Income', data=df)
plt.title("Income vs Product")
plt.show()
```



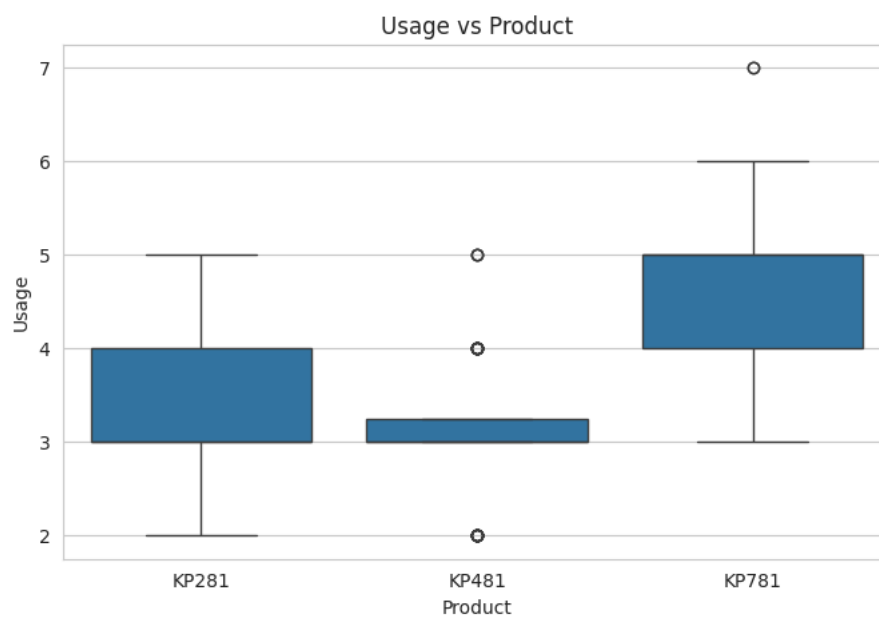
Miles vs Product

```
plt.figure(figsize=(8,5))
sns.boxplot(x='Product', y='Miles', data=df)
plt.title("Miles vs Product")
plt.show()
```



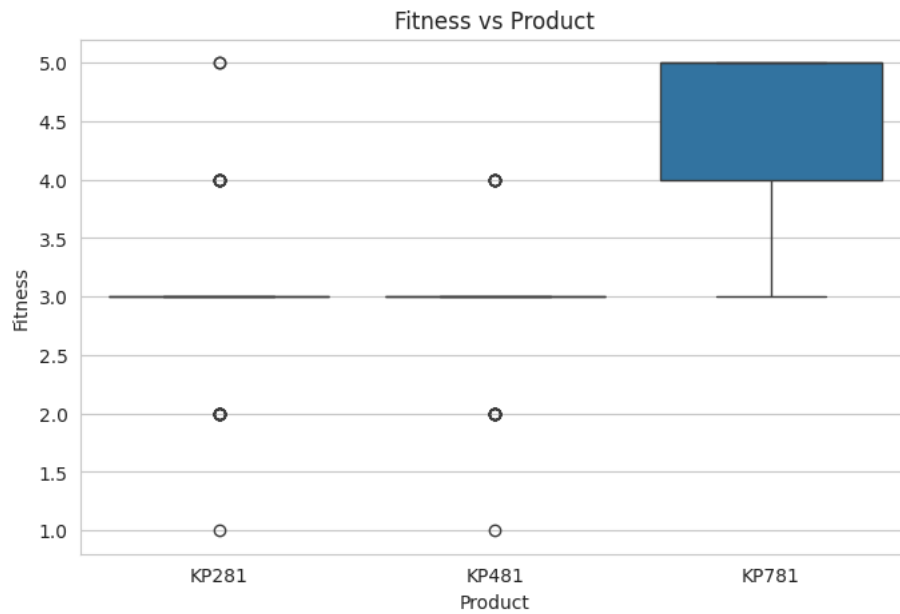
Usage vs Product

```
plt.figure(figsize=(8,5))
sns.boxplot(x='Product', y='Usage', data=df)
plt.title("Usage vs Product")
plt.show()
```



Fitness vs Product

```
plt.figure(figsize=(8,5))
sns.boxplot(x='Product', y='Fitness', data=df)
plt.title("Fitness vs Product")
plt.show()
```



Correlation Analysis

```
corr = df[numerical_cols].corr()
```

corr

	Age	Education	Usage	Fitness	Income	Miles
Age	1.000000	0.280496	0.015064	0.061105	0.513414	0.036618
Education	0.280496	1.000000	0.395155	0.410581	0.625827	0.307284
Usage	0.015064	0.395155	1.000000	0.668606	0.519537	0.759130
Fitness	0.061105	0.410581	0.668606	1.000000	0.535005	0.785702
Income	0.513414	0.625827	0.519537	0.535005	1.000000	0.543473
Miles	0.036618	0.307284	0.759130	0.785702	0.543473	1.000000

Next steps:

[Generate code with corr](#)[New interactive sheet](#)

```
plt.figure(figsize=(10,6))
sns.heatmap(corr,
            annot=True,
            cmap='coolwarm',
            fmt='.2f')

plt.title("Correlation Heatmap")
plt.show()
```